[This question paper contains 8 printed pages.]

Your Roll No

Sr. No. of Question Paper : 233 **I**

Unique Paper Code : 2932812302

Name of the Paper : Digital System Design

Name of the Course : **B. Tech – CSE (DSC-8)**

Semester : III

Duration : 3 Hours Maximum Marks : 90

Instructions for Candidates

1. Write your Roll. No. on the top immediately on receipt of this question paper.
2. Attempt five questions in all, including question no. 1 which is compulsory.
3. All questions carry equal marks.

1. (a) For each of the following Boolean expressions, specify the number of Implicants,

P.T.O.

Prime Implicants (PIs), and Essential Prime Implicants:

(i) $F = \Sigma_m(0,1,2,5,8,9,10,14,16,17,18,21)$

(ii) $F = \Sigma_m(2,3,5,7,11,13) + \Sigma_d(0,1,9,15)$

(iii) $F = \Sigma_d(0,1,9,15)$

(b) Design and implement a 32 × 1 MUX using only 16 × 1 and 2 × 1 MUX elements.

(c) Write a Verilog code to implement a 4 × 1 MUX. Write a testbench to test the working your code. There should be no syntax errors.

(d) Using only 2 × 1 MUX design the following gates:

(i) AND

(ii) OR

(iii) XOR (6+3+3+6 = 18)

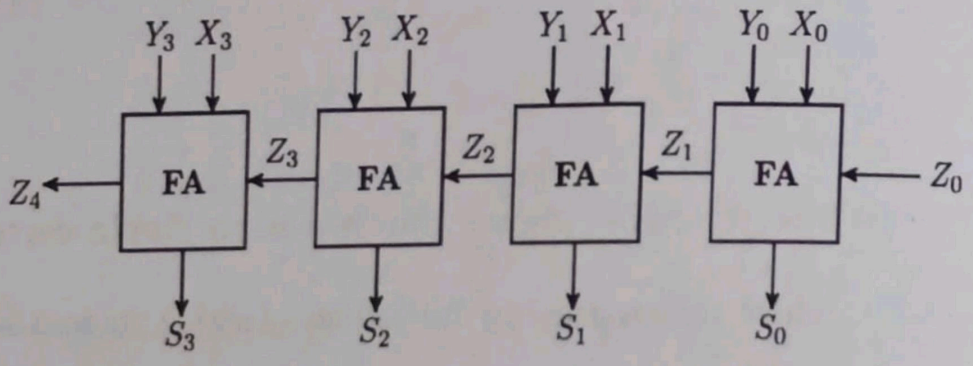
2. (a) Design a 4 – bit Excess-3 Adder Circuit.

(b) Design a JK counter that does through states 3,4,6,7 and starts again from 3. Is the counter self – starting? Modify the circuit such that whenever it goes to an invalid state it comes back to state 3. (9 × 2 = 18)

3. (a) The following figure shows a 4-bit ripple carry adder realized using full-adders (which in turn is realized using a combination of XOR, AND, and OR gates). The propagation delay of the XOR,

4

AND, and OR gates are 20 ns, 15 ns, and 10 ns, respectively. Assume all the inputs to the 4 – bit adder and Z_0 are initially reset to 0 (and it is kept 0 for a long time). Then at time $t = 0$, the inputs to the 4-bit adder are changed to $X_3X_2X_1X_0$=1100, $Y_3Y_2Y_1Y_0$=0110. Let t_0, t_1, t_2, t_3, and t_4 be the time when outputs S_0, S_1, S_2, S_3, and Z_4 respectively become stable (stop changing). Find the values of to through t_4.



(b) Design a MOD-16 ripple counter. How it can be converted in Decade Counter? Explain with suitable diagrams.

(9 × 2 = 18)

4. (a) Design a circuit to perform serial addition of two binary numbers using shift registers. One of the shift registers should be reused to store the resulting sum. A JK flip-flop should be used to manage the carry output, maintaining its current and next states during the addition process. Describe the working of the circuit and its components in detail, ensuring the design is efficient and meets the specified requirements.

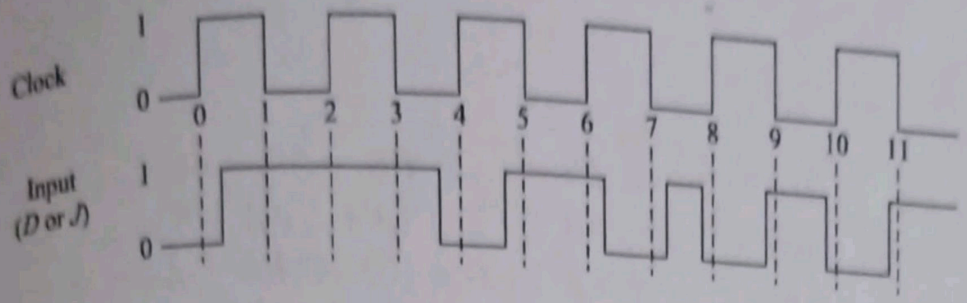
(b) In digital systems, it is often necessary to have circuits that can shift the bits of a vector by one or more-bit positions to the left or right. Design a

circuit that can shift a four-bit vector $W=w_3\ w_2\ w_1\ w_0$ one bit position to the right when a control signal Shift is equal to 1. Assume that the circuit outputs a four-bit vector $Y = y_3y_2y_1y_0$ and a one – bit value k, such that if $Shift = 1$ then $y_3 = 0, y_2 = w_3, y_1 = w_2, y_0 = w_1$, and $k=w_0$. If $Shift = 0$ then $Y = W$ and $k = 0$.

(9 × 2 = 18)

5. (a) Discuss the two techniques generally used to modify a latch into an edge-triggered flip-flop.

(b) Following clock and input waveforms are applied to D or J input of each of the following type of flip – flops. Sketch the output waveforms.



(i) Positive-edge-triggered D-type Flip Flop.

(ii) Positive-level-triggered D-type Flip Flop.

(iii) Negative-edge-triggered J-K Flip Flop (K = 1).

(iv) Master-slave J-K Flip Flop (K = 1).

(9 × 2 = 18)

6 (a) Simplify and prove the following Boolean expressions as indicated -

(i) Prove: $AB + \bar{A}C + BC = AB + \bar{A}C$

(ii) Prove: $AB + \bar{A}C = (A + C)(\bar{A} + B)$

(iii) Prove: $A\bar{B}C + B + B\bar{D} + AB\bar{D} + \bar{A}C = B + C$

(b) Expand A+BC $^{-}$+ABD $^{-}$+ABCD to minterms and maxterms. (9 × 2 = 18)